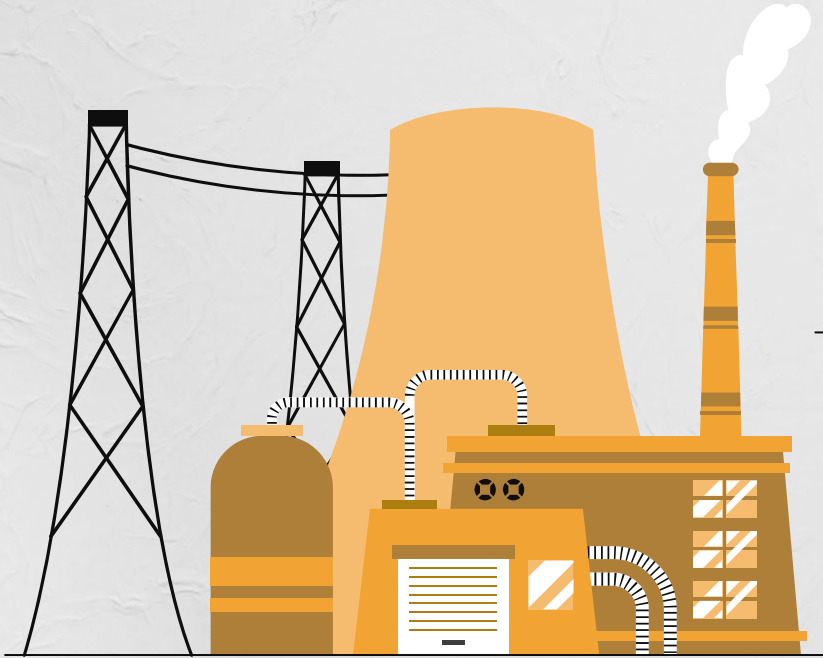


DS 4002
3/23/2026

When Will a Power Outage Occur?



Group 14: We KANT, but we can

Group leader: Tiannah Griffith
Alessia Natalucci, Nico Montoya, Katie Lai

Motivation

Electricity is a daily necessity

- Power outages disrupt businesses and daily life
- Predicting outages improves preparedness and response

Hypothesis:

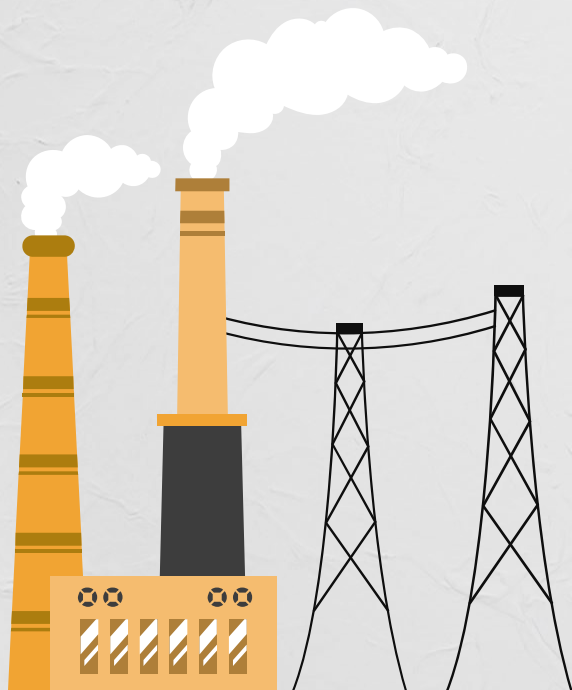
Our models will be able to predict the time of day of US power outages with a mean value error of less than 30 minutes.

Research Question:

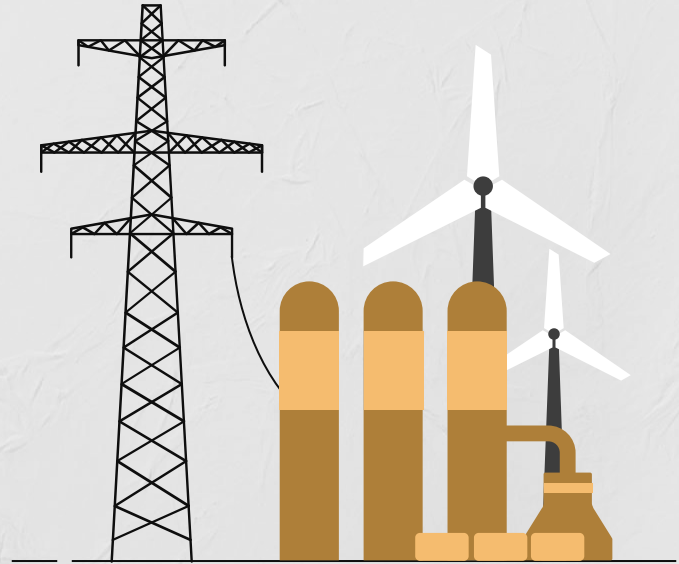
Can we develop a model that can predict the time of day of U.S. power outages in 2023 caused by various events?

Modeling Approach:

- Linear Regression Model
- Random Forest Model
- XGBoost Model



01 How can we acquire our data?

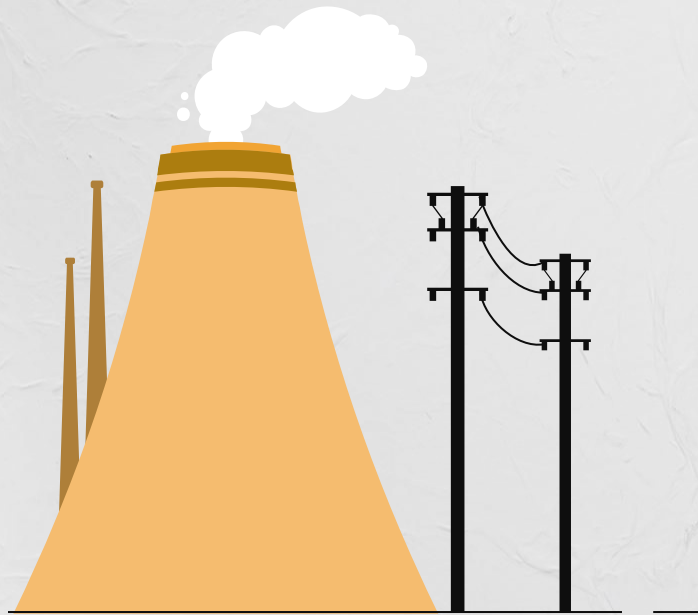


Event-correlated Outage Dataset in America from Data.gov

- Dataset: ~77,000 rows
- Source: [Data.gov](https://data.gov)
- Key features:
 - Event type
 - State
 - Start time → hour (target)
- Format: Time series power outage data

Column	Description
event_id	The count of outage in each state
Event Type	Cause of event
state	State in which outage occurs
county	County (in each state) in which outage occurs
start_time	The start time of the outage
duration	The length of time of the outage
end_time	The end time of the outage
hour	The time of day an outage began





02

**What was our
analysis plan?**



Analysis Plan

01

**Find Data +
Preprocessing**



02

**Data
Analysis**



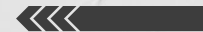
03

**Training
Strategy, Data
Splitting**



04

**Model
development**



05

**Model
Evaluation**



06

**Comparative
Analysis**



04

**What decisions
were
challenging?**



Tricky Analysis Decisions



Hyperparameters

Model took hours to run.
We had to carefully
decide how much we
tune the
hyperparameters



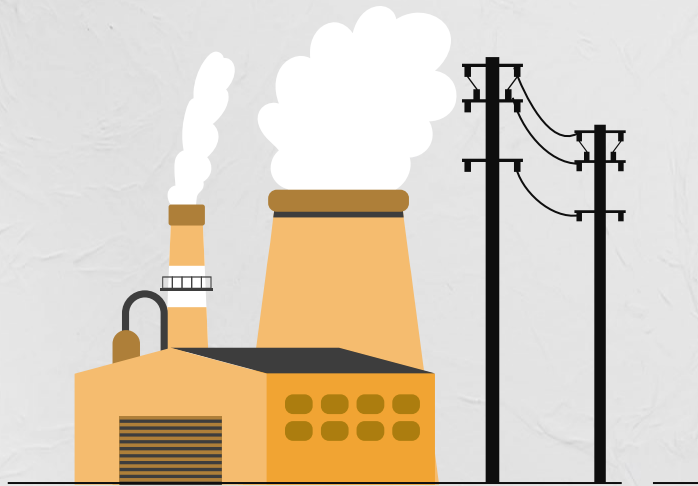
Duration vs Time

We wanted to predict
the duration of power
outages, but in that case
we wouldn't have been
using time series data



05

**Were there biases
or uncertainties?**



Bias And Uncertainty

Geographic Imbalance:

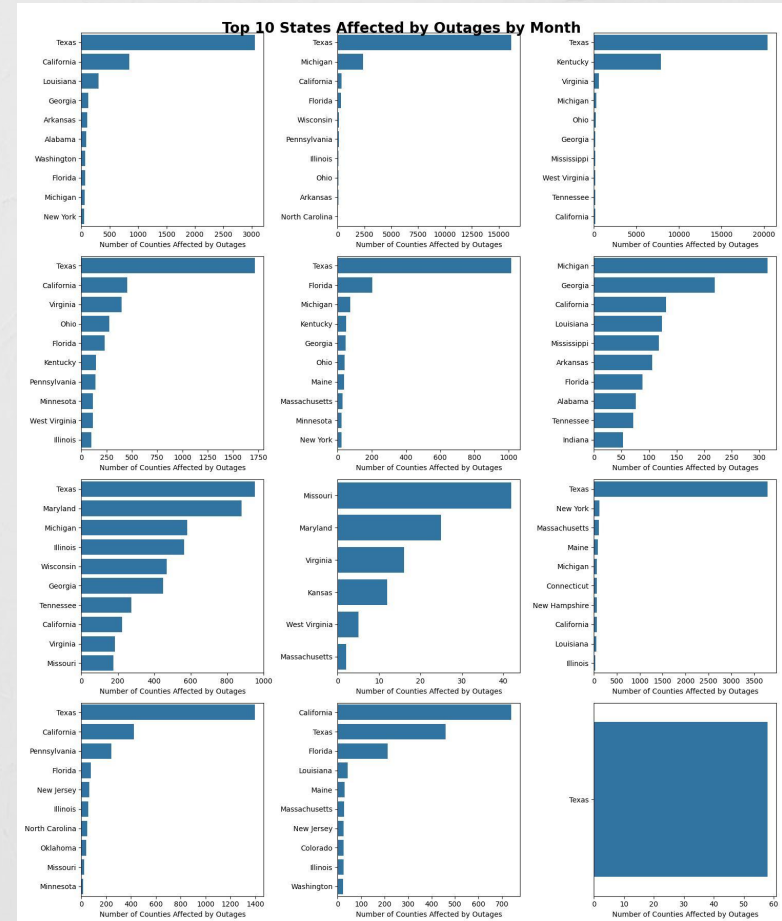
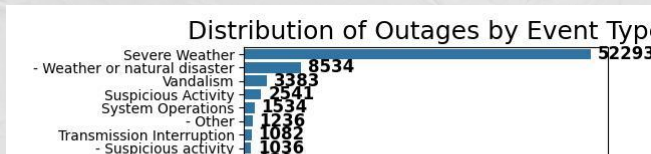
Large proportion of outages occurred in Texas across the months

- Geographic imbalance
- Model may overfit to Texas patterns

Event type Imbalance:

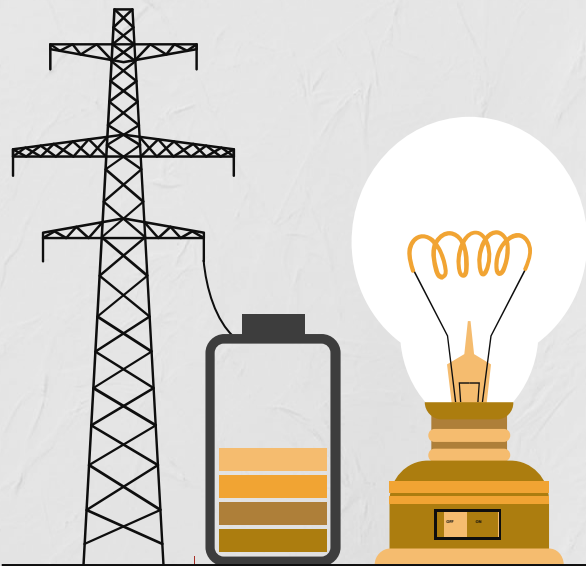
Majority of outages in the dataset are caused by weather-related events

- Dataset is skewed toward one type of outage cause
- Model may learn patterns that are specific to weather events only



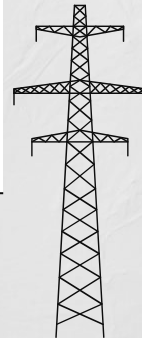
06

What we found...



Results – Before and After Hyperparameter Tuning

Model	R2 Before	R2 After	MAE Before	MAE After	RMSE Before	RMSE After
Linear Regression	0.166795	0.166795	5.456059	5.456059	6.658205	6.658205
Random Forest	0.790752	0.790984	1.508383	1.507656	3.336662	3.334807
XGBoost	0.535245	0.558044	3.830061	3.737221	4.972713	4.849206



Our Next Steps

Explore Feature importance



Refine Prediction Target: Time Bins



**How could predictions improve
response time and resource
allocation?**

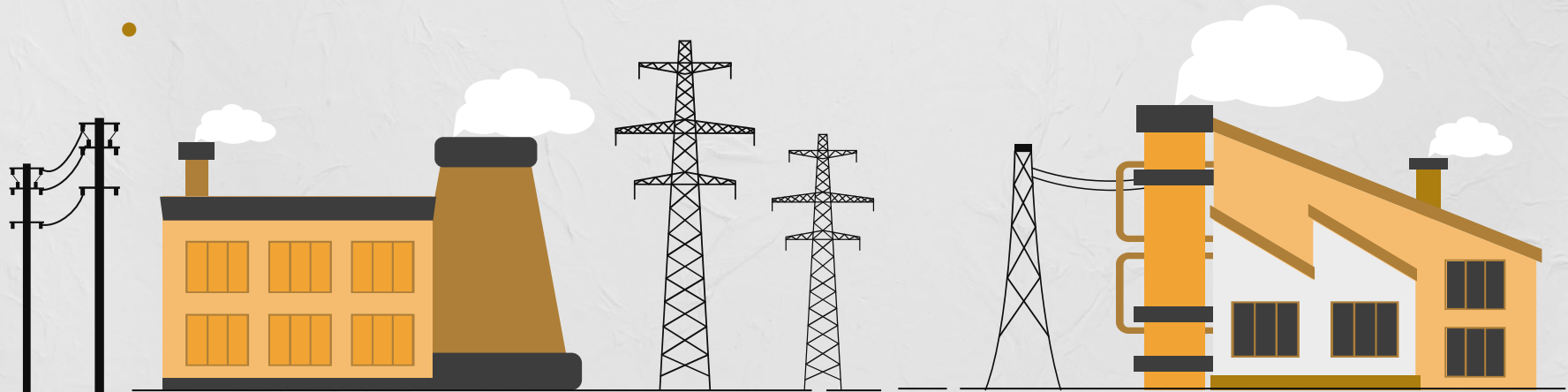


**Are outages predictable based on
location and event type?**



References

- [1] “Emergency electrician response times: For different situations,” ShockFree Electrical, <https://shockfreeelectrical.com/blog/emergency-electrician-response-times-for-different-situations/> (accessed Feb. 25, 2026).
- [2] “Data.gov,” *Data.gov*, 2025. <https://catalog.data.gov/dataset/event-correlated-outage-dataset-in-america> (accessed Feb 25, 2026).
- [3] M. Mohan, T. Ayyalasomayajula, Santhosh Bussa, and S. Ayyalasomayajula, “Forecasting Home Prices Employing Machine Learning Algorithms: XGBoost, Random Forest, and Linear Regression,” *Journal of Engineering & Technology Advancements*, vol. 1, no. 1, pp. 2583–2646, doi: <https://doi.org/10.56472/25832646/JETA-V1I1P114>.



Thanks!

Any questions?

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